byteBuilt 1.0

INNOVATE. IDEATE. INSPIRE.

Organized by byteXL & CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

SANKET × CivicLens

"Every report is a signal. Together, they reveal the real problem."

AI-Powered Civic Signal Intelligence Platform

Team InfraNova | byteBuilt 1.0 — byteXL & Chandigarh University

CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

byteBuilt 1.0
INNOVATE. IDEATE. INSPIRE.

PROBLEM STATEMENT & TEAM DETAILS

UNDERSTAND TODAY | SOLVE TOMORROW | CLEANER CITIES

THE PROBLEM

Cities don't lack complaints.
They lack intelligence.

01 **FRAGMENTED REPORTING**
Urban civic issues are reported as isolated complaints.

02 **NO UNIFIED INTELLIGENCE**
Authorities lack a unified intelligence layer to combine citizen observations and identify recurring incidents.

03 **DIFFICULT PRIORITIZATION**
Risk, impact and urgency are difficult to assess consistently, making intervention reactive.

04 **UNCERTAIN RESOLUTION**
Authorities need better ways to verify whether reported infrastructure issues have actually been resolved.

FROM CIVIC CHALLENGES TO REAL-WORLD IMPACT

TEAM INFRANOVA

Problem Statement
AI-Powered Civic Infrastructure Intelligence & Prevention Platform

Ridhi
Backend, Database & Data Analytics

Hiteshi
AI/ML & Computer Vision

Aradhya
UI/UX, Product Design & Presentation

Kiran
Frontend & Web Development

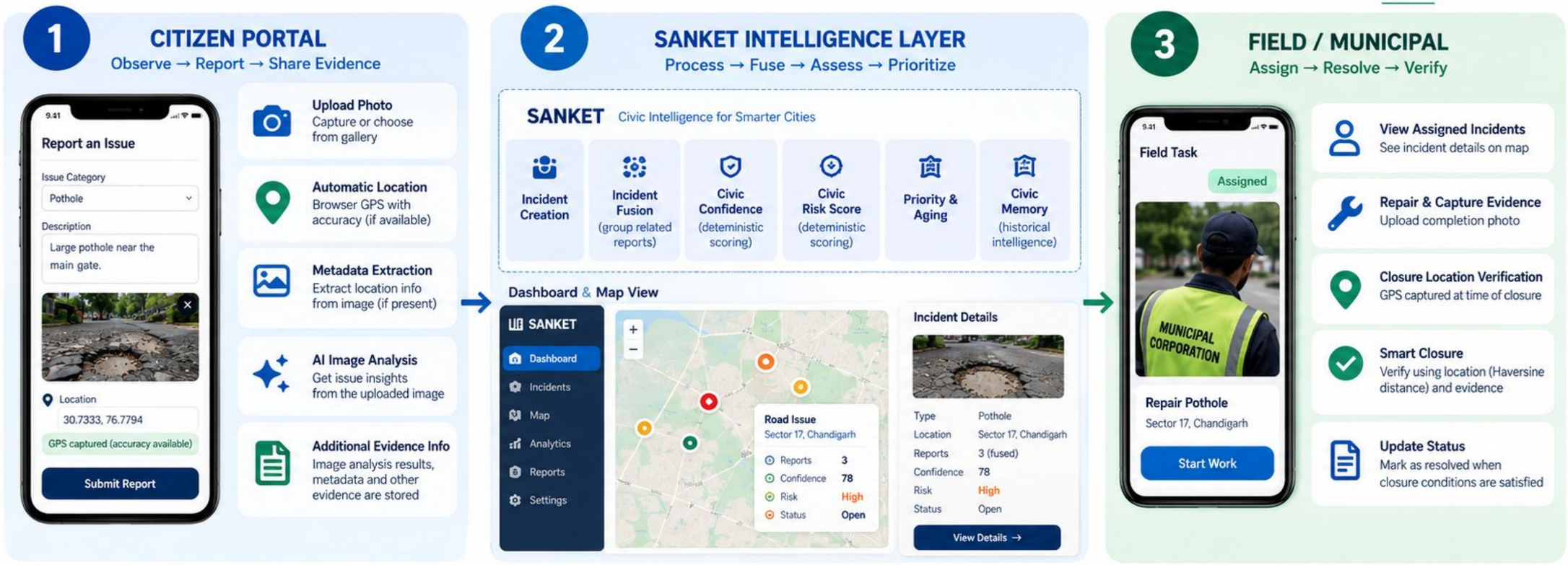
BACKEND + AI/ML + FRONTEND + PRODUCT DESIGN
A cross-functional team covering the complete product stack.

PEOPLE | TECHNOLOGY | INTELLIGENCE | CLEANER CITIES

Sources are publicly verifiable and used to ground the solution in established civic, technological and government contexts.

SANKET × CivicLens

Proposed Solution: One Photo. Zero Forms. Smarter Civic Action.



SANKET is the intelligence layer that interprets civic signals, while **CivicLens** transforms those signals into **actionable civic intelligence.**

> **CivicLens** doesn't just collect complaints — **SANKET** tells cities if an issue is real, recurring, risky, and waiting too long.

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

TRADITIONAL CIVIC REPORTING
Common Challenges

- Multiple forms and manual data entry
- Reactive, not proactive
- Duplicate complaints for the same issue
- Limited evidence and context
- Uncertain prioritization
- Slow and opaque resolution tracking

SANKET × CIVICLENS
A Smarter, Evidence-Driven Approach

- Photo-based reporting with automatic location capture
- Metadata extraction from images and reports
- Incident fusion to group related reports
- Deterministic Civic Confidence & Civic Risk scoring
- Priority aging to highlight long-pending issues
- Smart Closure Match using evidence and location verification

FROM REPORTS TO REAL IMPACT

01 CIVIC CONFIDENCE
How certain are we about this incident?

Deterministic scoring based on available report data and corroboration.

Score shown in dashboard (no arbitrary values).

02 INCIDENT FUSION
Group related reports automatically.

Identifies multiple reports referring to the same underlying issue.

Reduces duplicate complaints and provides a unified incident view.

03 CIVIC MEMORY
Learn from past incidents.

Maintains historical incident data and patterns.

Helps identify recurring issues and supports better prioritization.

04 PRIORITY AGING
Highlight issues that are waiting longer.

Uses time factor along with confidence, risk and other signals.

Long-pending and high-impact issues surface with higher priority.

05 SMART CLOSURE MATCH
Verify before closing.

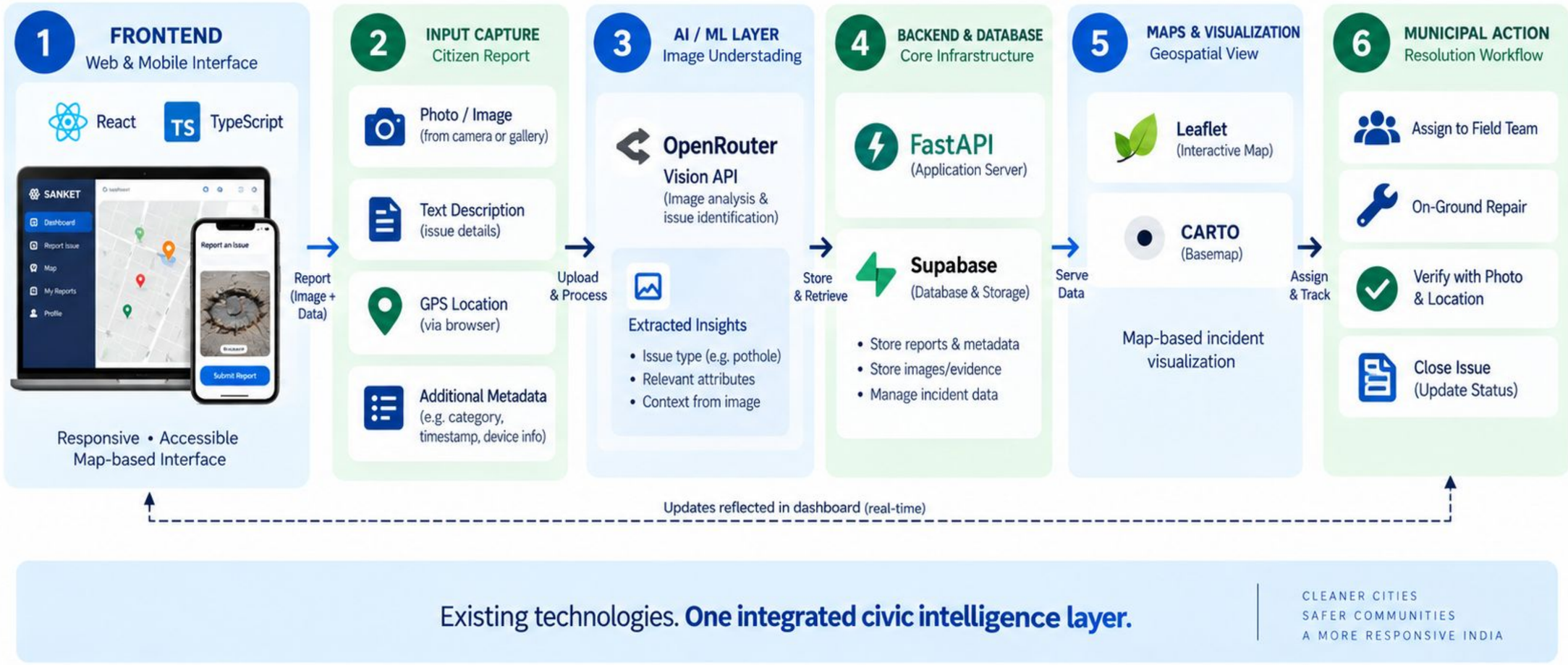


Matches repair photo and verifies location (Haversine distance) before closure.

SANKET turns civic reports into evidence-backed priorities.

CivicLens captures the signal.
SANKET verifies, remembers, scores and prioritizes it.

TECHNICAL APPROACH



Existing technologies. **One integrated civic intelligence layer.**

CLEANER CITIES
SAFER COMMUNITIES
A MORE RESPONSIVE INDIA

FEASIBILITY & VIABILITY

ECONOMIC & OPERATIONAL VIABILITY (COMPARATIVE VIEW)

Aspect	Existing Portals (e.g. Local / Govt.)	With SANKET	Expected Impact
Citizen Effort	Multiple forms, follow-ups, often manual	One evidence-backed report with tracking	Less citizen time and follow-up effort
Administrative Workload	Duplicate complaints consume staff attention	Related reports consolidated into one incident	Less duplicated administrative work
Issue Identification	Delays in identifying critical infrastructure problems	Earlier prioritization of important issues	Potentialy lower downstream repair costs
Closure Verification	Often manual and unverified, leading to repeat complaints	Photo + location evidence supports closure verification	Fewer unnecessary re-checks and field visits
Long-Term Cost to City	Delayed action can increase damage and repair costs	Timely attention can prevent larger issues	Reduced risk of avoidable infrastructure costs

SOCIETAL IMPACT

Benefits for all stakeholders

For Citizens

- Fewer repeated forms and follow-ups
- Less uncertainty about issue status
- Better evidence trail for reported problems
- A more convenient and trustworthy experience

For Municipal Teams

- Less time spent sorting duplicate or incomplete complaints
- Better visibility of issues requiring attention
- More structured evidence for field verification
- Ability to focus on high-priority issues

For the City

- Faster identification of recurring problems
- Reduced possibility of unresolved issues being repeatedly reported
- Earlier action can help prevent avoidable property damage, traffic disruption and safety risks
- Cleaner, safer and more liveable cities

ECONOMIC LOGIC (REALISTIC & DEFENSIBLE)

SANKET reduces avoidable operational effort by consolidating evidence, reducing duplicate handling and supporting earlier prioritization. Earlier identification of infrastructure problems can also reduce the likeilhood of larger downstream repair and societal costs.

> **“Less citizen nuisance → less duplicate administrative effort → earlier attention → potentially lower cost of delayed civic action.”**

IMPACT & BENEFITS

DIFFERENT USERS. A COMMON GOAL.

Turning everyday civic issues into actionable solutions.

CLEANER CITIES | SAFER COMMUNITIES | MORE ACCOUNTABLE GOVERNANCE | IMPROVED QUALITY OF LIFE



CITIZENS

"Report. Track. See Change."

Why they will use it?

- Report civic issues easily with a photo.
- No complex forms or repeated submissions.
- Track status and see progress.
- Have confidence that their report reaches the right department.

Impact for Citizens

- Less time and effort
- More transparency and trust
- Reduced follow-ups and frustration
- A cleaner and safer neighbourhood

"A simpler way to be heard, and to see change in my city."



MUNICIPAL AUTHORITIES

"From Reports to Results."

Why they will use it?

- Receive well-structured, location-tagged reports with necessary details.
- Identify and manage duplicate reports more efficiently.
- Prioritize issues based on evidence and severity.
- Monitor progress and ensure accountability across teams.
- Use data for better planning and resource allocation.

Impact for Municipal Authorities

- Reduced manual workload
- Better visibility and decision-making
- More efficient use of resources
- Structured data for long-term planning

"Helps us respond faster, work smarter, and serve citizens better."



FIELD WORKERS

"Clear Tasks. Faster Action."

Why they will use it?

- Get clear task assignments with location and issue details.
- See relevant photos and descriptions before visiting the site.
- Update status with on-ground photos after resolution.
- Reduce unnecessary visits with better information.
- Focus on genuine, high-priority issues.

Impact for Field Workers

- Less time spent on unclear or incomplete tasks
- Accurate location and issue details
- Easier status updates and closure
- Safer and more organized field operations

"The right information, in the right place, at the right time."



FOR SOCIETY & THE CITY

"Together for a Better Tomorrow."

Why it matters?

- Quicker resolution of civic issues.
- Cleaner, safer and more liveable neighbourhoods.
- Greater accountability and transparency in civic operations.
- Data-driven insights for long-term infrastructure planning.
- Strengthens citizen participation in governance.

Broader Impact

- Cleaner environment
- Improved public safety
- More responsive governance
- Better infrastructure planning
- Higher quality of life for all

"Civic responsibility today, stronger cities tomorrow."

BUSINESS MODEL

1 TARGET CUSTOMERS

PUBLIC INSTITUTIONS | COMMUNITIES | ENTERPRISES

- Municipal Bodies
- University Campuses
- Residential Communities
- Industrial Parks
- Smart Campuses
- Infrastructure & Facility Management Organizations

CITIZENS
Report issues from anywhere

→ **CivicLens**
Free for Citizens
Capture • Report • Track

→ **Civic Signals**
Real-time, structured data from the ground

→ **SANKET Intelligence**
For Organizations
Analyze • Prioritize • Enable Action

→ **ORGANIZATIONS**
Better decisions.
Cleaner cities.

B2G + B2B · Free for Citizens. Intelligence for Organizations.

3 HOW WE MONETIZE

SUSTAINABLE | SCALABLE | IMPACT-DRIVEN

01	PILOT	Small-scale pilot for one campus / ward / zone
02	PLATFORM LICENSE	Annual SaaS licensing for municipal deployment
03	ENTERPRISE DEPLOYMENT	Custom integration and deployment for larger bodies
04	ANALYTICS MODULE	Advanced infrastructure analytics add-on
05	API / WHITE-LABEL	Integration with existing grievance systems

4 YEAR 1: VALIDATE → PILOT → IMPROVE → COMMERCIALIZE

FROM PROTOTYPE TO REAL-WORLD IMPACT

01	VALIDATE Months 1 – 3	Build MVP, test citizen reporting & incident fusion	→
02	PILOT Months 4 – 6	Campus/community pilot, gather feedback, improve	→
03	IMPROVE Months 7 – 9	Improve Civic Memory, hotspot intelligence & Priority Aging	→
04	COMMERCIALIZE Months 10 – 12	Small paid pilots, organization dashboard, analytics	

VALIDATE → PILOT → IMPROVE → COMMERCIALIZE
B2G + B2B · Free for Citizens. Intelligence for Organizations.

SANKET × CivicLens

RESEARCH & REFERENCES

1 KEY REFERENCE PLATFORMS

These government initiatives and platforms were studied to understand existing systems, citizen needs and best practices in urban governance.

CPGRAMS

सत्यमेव जयते

Centralized Public Grievance Redress and Monitoring System
Government of India

National platform for lodging and tracking grievances across Ministries and Departments. Useful to understand grievance workflows, tracking mechanisms and citizen expectations.

https://pgportal.gov.in/

Citizen Grievance Redressal

Smart Cities Mission

Smart Cities

Ministry of Housing and Urban Affairs
Government of India

Provides insights into the use of technology, GIS and data-driven approaches for urban infrastructure management and citizen services in Indian cities.

https://smartcities.gov.in/

Urban Governance & Smart Infrastructure

Swachh Bharat Mission

स्वच्छ भारत
एक कदम स्वच्छता की ओर

Ministry of Housing and Urban Affairs
Government of India

Guidelines and resources on sanitation, waste management and citizen participation. Relevant for understanding civic cleanliness challenges and community-driven solutions.

https://swachhbharatmission.ddws.gov.in/

Sanitation & Clean Cities

Digital India Programme

Digital India
Power To Empower

Ministry of Electronics and Information Technology
Government of India

National vision for Digital Governance, citizen services and technology-enabled public service delivery. Provides broader context for digital solutions in governance.

https://www.digitalindia.gov.in/

Digital Governance & Citizen Services

2 HOW THESE REFERENCES INFORMED SANKET

Key learnings from existing government platforms that shaped our approach.

Citizen-Centric Design
Learned the importance of simple and accessible grievance reporting from CPGRAMS and Digital India.

Use of Technology
Understood the value of GIS, data-driven planning and real-time monitoring from the Smart Cities Mission.

Focus on Civic Cleanliness
Recognized key sanitation and waste management challenges and the role of citizen participation from Swachh Bharat Mission.

Scalable Governance
Derived insights on building scalable, inclusive and transparent public service systems from the Digital India Programme.

Learning from real initiatives. Building for real impact.

CLEANER CITIES
SAFER COMMUNITIES
A MORE RESPONSIVE INDIA

CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

byteBuilt 1.0
INNOVATE. IDEATE. INSPIRE.

THANK YOU

SANKET × CivicLens

Every report is a signal. Together, they reveal the real problem.



InfraNova

Ridhi · Hiteshi · Aradhya · Kiran

AI-POWERED CIVIC INFRASTRUCTURE INTELLIGENCE & PREVENTION PLATFORM